VISION VIGILANTS
IBVAP SURVEILLANCE

TECHNICAL APPROACH

INTELLIGENT BORDER VIDEO ANALYTICS PLATFORM

SMART INDIA HACKATHON 2026

VIDEO INPUT 01

4 camera sources
Own perimeter footage, highway dashcam, border wall, RTSP template

OpenCV capture
File, webcam or **RTSP** behind one interface, so the pipeline never knows which

Frame gate
Every 2nd frame analysed. Halves compute and loses nothing: a boundary cannot be crossed inside one frame

YAML configuration
`application · cameras · zones · threat_rules`

PERCEPTION 02

YOLOv8n detection
6 classes of 80. **640 px**, confidence 0.35, IoU 0.50

ByteTrack identity
Published after **3 hits**, coasts 5 frames, re-identified for 30

YOLOv8n-pose
17 keypoints on a 192 px crop of the person, not the whole frame, so a distant figure keeps resolution

Pose history
A time series per track: posture becomes something with a direction and a speed

ANALYSIS — THE BRAIN 03

Movement, in body heights
Every distance and speed is measured in the person's **own body height**, never in pixels. Walking scores 0.45 and running 1.9 whether they are near the camera or far from it — which is why one set of thresholds works on all four cameras and no per-camera calibration exists

Behaviour classifier
`WALKING · RUNNING · BENT_WALK · BENT_RUN · CROUCHING · JUMPING · FALLING · STANDING`

Boundary interaction
`NONE → APPROACHING → AT_BOUNDARY → INTERACTING → CROSSED`

A crossing is armed, then confirmed, so a person walking the fence is not a person going over it

Isolation Forest anomaly
scikit-learn, trained on **this camera's own** normal movement. Refits every 25 samples

JUDGEMENT 04

Threat engine
16 weighted factors in a YAML file, not buried in code, so the reasoning can be read and argued with

Four levels
`LOW 0–24` `MEDIUM 25–49`
`HIGH 50–74` `CRITICAL 75+`

Incident & alert
Deduplicated, so a threat lasting 100 frames is **one** incident and one alert, not a hundred

Evidence capture
Full frame and crop written at MEDIUM and above, indexed to the incident that caused them

OPERATOR DELIVERY 05

FastAPI — one process
Serves the API **and** the compiled dashboard on port 8000. No second server, no CORS, nothing to go stale

React 19 command wall
4 live feeds, threat timeline, evidence strip. Boxes are drawn **server-side**, so the screen cannot show a detection the pipeline did not make

SQLite — 16 tables
WAL mode, a file not a server. Evidence as files plus an index, never blobs

Access control
scrypt hashing, 2 roles, enforced by the **backend** on every request — not by hiding React routes

ANPR BRANCH
Runs only inside a vehicle box. Optional: absent, it says so.

Plate region proposal
Blackhat morphology raises dark-on-light text, a Sobel pass finds the dense vertical edges characters make

EasyOCR
Measured on our own footage: **13.6%** hit rate against Tesseract's 1.8%

Vote per vehicle
Reads grouped by tracked vehicle and resolved by confidence weight — one car, one answer, not one row per frame

RTO origin decoder
41 state codes, 998 district codes, standard library only. `PB39N0203 → Rajpura, Punjab` · `HR26BQ0008 → Gurgaon, Haryana`. A code not in the table is reported as missing, never guessed

EVIDENCE & AUDIT TRAIL

Evidence, not screenshots
Full frame and crop written at capture time and indexed to the incident, camera, track and timestamp that produced them. Served only through an authenticated route, never by path

Tokens never stored usable
A session token exists only as its SHA-256 fingerprint in the database, so a copy of the database cannot be replayed as a login

Audit log
Every sign-in, refusal, user change and camera command is recorded with who, what and when — including the 403s that were denied

DEPLOYMENT & OPERATION

Runs on one laptop
No server, no container, no cloud account. SQLite is a file and the models are three files totalling 19.6 MB, so the whole system fits on the machine that runs it

One double-click
A macOS app bundle stops any stale server, waits for the API to actually answer rather than guessing, and reports failure in a dialog instead of silently not opening

Degrades honestly
Missing OCR engine, empty database and unreachable backend are three different messages, because they need three different fixes. Nothing is ever shown as a blank

One process. No cloud. Every number on screen is one the pipeline computed.

460	29	27	19.6 MB	0
TESTS PASSING	ANALYSIS MODULES	API ENDPOINTS	MODEL WEIGHTS	CLOUD SERVICES